

Project ORION: Quantum Encryption Protocols

CONFIDENTIAL: LEVEL 5 CLEARANCE REQUIRED

1. INTRODUCTION

Project ORION was initiated in 2025 to develop a next-generation quantum encryption standard for deep-space communications.

Unlike traditional RSA or AES encryption, ORION utilizes quantum entanglement to ensure that any interception of the signal immediately destroys the data, alerting the sender to the breach.

2. PROTOCOL SPECIFICATIONS

- Encryption Algorithm: Quantum Lattice Key Distribution (QLKD)
- Qubit Transmission Rate: 4.5 Gigabits per second
- Entanglement Range: 15,000,000 Kilometers
- Error Correction: Adaptive AI Neural Network (Nexus-Core v4.2)

3. KNOWN VULNERABILITIES (FOR INTERNAL REVIEW ONLY)

In recent simulation tests, a vulnerability was discovered when the QLKD system is exposed to extreme magnetic interference (e.g., solar flares). The error correction AI can occasionally misinterpret the magnetic noise as an intentional interception, causing a false-positive data purge.

4. DIRECTIVES

Engineers are required to patch the Nexus-Core v4.2 AI logic by incorporating solar-flare telemetry data into the error correction heuristic by Q4 2026.